Reg.No : .....

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2026

FOURTH SEMESTER

Value Addition Course (VAC) – MG4VACECT200- Environmental monitoring using sensors

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember (K), Understand (U), Apply(A), Analyse(An), Evaluate (E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Question

Answer all questions

Each question carries **1**

marks

1. Environmental monitoring mainly involves the measurement of…….. [CO1][U]

A) Economic growth

B) Environmental parameters

C) Industrial profit

D) Population growth

2. The greenhouse effect is mainly caused by [CO2][A]

A) Reflection of sunlight

B) Trapping of heat by gases

C) Ozone depletion

D) Wind circulation

3. MQ135 can detect gases such as [CO3][C]

A) Oxygen and Nitrogen
B) CO_2, NH_3, smoke and benzene
C) Helium only
D) Hydrogen only

4. Which of the following is NOT an environmental parameter? [CO1][U]

A) Air quality
B) Water quality
C) Soil moisture
D) Stock market index

5. Which gas is the major contributor to the greenhouse effect? [CO2][A]

A) Oxygen
B) Nitrogen
C) Carbon dioxide
D) Argon

6. The analog output pin of MQ135 provides [C03][A]

A) Digital ON/OFF signal
B) Pulse width signal
C) Voltage proportional to gas concentration
D) Constant voltage

7. Which of the following is a regulatory measure to control air pollution? [CO4][C]

A) Installing air purifiers at homes
B)Imposing emission standards on factories and vehicles
C)Planting more trees
D)Educating the public about air pollution

8. Monitoring air pollutants such as CO_2 and NOx helps in [CO1][U]

A) Increasing industrial output
B) Predicting weather
C) Controlling air pollution
D) Increasing traffic

9. Methane is considered more harmful than CO_2 because it [CO2][A]

A) Is heavier
B) Has higher global warming potential
C) Is radioactive
D) Reduces oxygen

10. Which pin of MQ135 gives analog output? [CO3][C]

A) VCC
B) ground
C) A0
D) D0

11. MQ135 sensor is widely used with [CO3][C]

A) Raspberry Pi only
B) PLC
C) Arduino
D) Laptop

12. What is the main purpose of air filtering systems in cities? [CO4][C]

A)To reduce noise pollution
B)To trap harmful pollutants and particulate matter from the air
C)To increase vehicle speed
D)To promote industrial growth

13. Vehicle emission control measures include: [CO4][C]

A) Using catalytic converters
B) Reducing vehicle density
C) Switching to cleaner fuels
D) All of the above

14. Continuous environmental monitoring helps governments in [CO1][U]

A) Increasing taxes
B) Policy formulation
C) Entertainment planning
D) Advertising

15. Monitoring of PM2.5 and PM10 is related to [CO1][U]

A) Water pollution
B) Noise pollution
C) Air pollution
D) Soil pollution

16. Which greenhouse gas is released mainly from refrigeration and air conditioners? [CO2][U]

A) CO_2
B) CH_4
C) CFCs
D) N_2

17. An increase in greenhouse gases leads to [CO2][U]

A) Global cooling
B) Climate stability
C) Global warming
D) Reduced rainfall

18. Air Quality Index (AQI) is used to [CO2][U]

A) Measure rainfall
B) Indicate air pollution level
C) Measure soil fertility
D) Predict earthquakes

19. AQI mainly represents the concentration of [CO2][U]

A) Water pollutants
B) Noise levels
C) Air pollutants
D) Soil minerals

20. While reading analog data from MQ135 using Arduino, the sensor output is connected to [CO2][U]

A) Digital pin
B) PWM pin
C) Analog input pin
D) Reset pin

21. In the block diagram of MQ135 interfacing, the correct sequence is [CO2][A]

A)Sensor → Display → Arduino → Power supply
B) Arduino → Sensor → Serial monitor → Power supply
C) MQ135 → Arduino → Serial Monitor
D) Power supply → Serial monitor → Sensor

22. A weather station is mainly used to [CO4][C]

A) Predict earthquakes
B) Monitor atmospheric parameters
C) Control pollution
D) Measure soil fertility only

23. Environmental monitoring plays a major role in addressing [CO1][U]

A) Political issues

B) Global environmental issues

C) Cultural conflicts

D) Economic inflation

24. Which environmental issue is mainly monitored using greenhouse gas measurements? [CO2][A]

A) Soil erosion

B) Climate change

C) Noise pollution

D) Water scarcity

25. Water quality monitoring is important to prevent [CO2][A]

A)Global warming

B) Acid rain

C) Waterborne diseases

D) Deforestation

26. Which technology is widely used today for real-time environmental monitoring? [CO2][A]

A)Manual observation

B) Typewriter

C) IoT and sensors

D) Telephone exchange

27. What is the primary role of IoT in environmental monitoring? [CO1][U]

A) Increasing industrial production

B) Real-time data collection and remote monitoring

C) Reducing the number of sensors used

D) Eliminating the need for weather stations

28. Which sensor is used to measure temperature? [CO2][A]

A) Gas sensor
B) Temperature sensor
C) Smoke sensor
D) Pressure sensor

29. Why is environmental monitoring important? [CO1][U]

A) To increase industrial production
B) To detect and control environmental problems
C) To reduce the use of sensors
D) To avoid environmental laws

30. Environmental monitoring helps mainly to: [CO1[U]

A) Increase pollution
B) Identify changes in environmental conditions
C) Stop all industrial activities
D) Replace natural resources

31. Which parameter is commonly monitored in the environment? [CO3][C]

A) Internet speed
B) Air quality
C) Mobile battery level
D) Screen resolution

32. Environmental monitoring data is useful for: [CO3][C]

A) Making environmental decisions
B) Playing games
C) Online shopping
D) Watching movies

33. Which of the following is an environmental parameter? [CO3][C]

A) Temperature
B) Internet speed
C) Screen size
D) Phone battery

34. Real-time environmental monitoring helps in: [CO3][C]

A) Delayed decision making
B) Quick response to environmental hazards
C) Increasing fuel consumption
D) Ignoring environmental issues

35.Air quality monitoring mainly measures**:** [CO3][C]

A) Sound level
B) Gas pollutants
C) Soil nutrients
D) Water level

Part B

Short Essay Questions
Answer any 3 questions
Each question carries 5 marks

36. Describe various environmental sensors used for monitoring temperature, humidity, and air quality, including carbon monoxide, smoke, methane, and ozone sensors, with their working principle . [CO2][A]

37. Explain the concept of the Air Quality Index (AQI) and its importance. [CO2][A]

38. Explain the importance of environmental monitoring in addressing global environmental issues, giving real-world giving real world examples. [CO1][U]

39. Explain the concept of a weather station and its functions. [CO4][C]

40. Discuss the relationship between vehicle density and air pollution. How does increased traffic contribute to air quality deterioration, and what measures can be taken to reduce its impact? [CO3][C]